Let $x_1, \cdots, x_n \overset{i.i.d.}{\sim} P$ where P is supported on $\mathbb{R}^d$ with $\|x\|_2, \|\beta_0\|_2 \leq B$ almost surely. Let $X = (x_1, \cdots, x_n) \in \mathbb{R}^{n \times d}$ and $y = X\beta_0 + \varepsilon$, with ε being centered and sub-Gaussian. Throughout we omit poly-logarithmic terms in $\widetilde{O}(\cdot)$ and $\lesssim$ notations. When we say "with high probability", it means with probability $1 - O(T^{-4})$ or higher degrees.

Let $V_0 = \mathbb{E}_P[xx^\top]$ and $\widehat{V} = \sum_{i=1}^n x_i x_i^\top = \frac{1}{n} X^\top X$. For $\lambda > 0$, let $V_\lambda = V_0 + \lambda I$ and $\widehat{V}_\lambda = \widehat{V} + \lambda I$ be Ridge terms. The Ridge regression is defined as

$$\widehat{\beta} = \widehat{V}_\lambda^{-1} \frac{1}{n} X^\top y.$$

The objective is to construct UCB $\Delta(\cdot)$ so that it satisfies the following properties:

1. With high probability $|x^\top(\widehat{\beta} - \beta_0)| \leq \Delta(x)$ for all $x \in \mathcal{X}$, where $\mathcal{X}$ contains T vectors in the support of P;
2. $\mathbb{E}_P[\Delta(x)] \leq \Psi$ where Ψ is on the order of $\widetilde{O}(1/\sqrt{n})$.

Note that we don't require $\Delta(x)$ to be valid *uniformly* on the support of P, but it has to be valid pointwise with high probability so that a union bound of T vectors won't hurt.

0.1 Successive elimination

It is possible to use such an UCB to do successive elimination, without requiring $\lambda_{\min}$ conditions. Algorithm operates as below:

1. For epochs $\tau = 1, 2, 3, \cdots$ each of length $n_\tau = 2^\tau$, do the following:
   (a) Obtain $\widehat{\beta}_\tau^1$, $\widehat{\beta}_\tau^2$ and $\Delta_\tau^1(\cdot)$, $\Delta_\tau^2(\cdot)$ from all samples in the previous epoch;
   (b) For all t in epoch τ, observe $x_t \sim P$ and do the following:
      i. If $x_t^\top \widehat{\beta}_s^1 - \Delta_s^1(x_t) > x_t^\top \widehat{\beta}_s^2 + \Delta_s^2(x_t)$ for any $s \leq \tau$ then take action 1, and similarly in the other direction take action 2;
      ii. Otherwise, take action 1 or 2 with half probability each.

Analysis. With high probability the optimal action is never eliminated. So regret is only incurred when actions are taken with half probability each. Define distribution $z \sim Q_\tau$ as the following: first sample $x \sim P$, then set $z = x$ if on x actions are taken with half probability each during epoch τ, and $z = 0$ otherwise, with the understanding that $\Delta_\tau^1(0) = \Delta_\tau^2(0) = 0$. Let P_τ^1, P_τ^2 be the distributions received by actions 1 and 2 from all samples in the previous epoch. We have the following observations:

1. $\mathbb{E}[\Delta_\tau^1(x)|P_\tau^1], \mathbb{E}[\Delta_\tau^2(x)|P_\tau^2] \lesssim \Psi_\tau \lesssim 1/\sqrt{n_\tau}$, where the last inequality is hiding poly(d) factors. This holds true as long as the two actions are balanced.
2. For any $x \in \text{supp}(Q_\tau) \backslash \{0\}$, $p(x|Q_\tau) \leq \max\{2\kappa_{\min}^{-1} p(x|P_\tau^1), 2\kappa_{\min}^{-1} p(x|P_\tau^2)\}$ where $\kappa_{\min} = \min\{P(a^*(x) = 1), P(a^*(x) = 2)\}$ because x is not eliminated during epoch $\tau - 1$ and therefore has half probability being assigned to each action.

Therefore, during epoch τ, the cumulative regret is

$$n_\tau \mathbb{E}[\Delta_\tau^1(z) + \Delta_\tau^2(z)|Q_\tau] \lesssim n_\tau(\mathbb{E}[\Delta_\tau^1(x)|P_\tau^1] + \mathbb{E}[\Delta_\tau^2(x)|P_\tau^2]) \lesssim n_\tau \times \frac{1}{\sqrt{n_\tau}} = \widetilde{O}(\sqrt{n_\tau}).$$

This leads to $\widetilde{O}(\sqrt{T})$ regret. Dimension dependency depends on Ψ and varies.

0.2 Direct analysis of Ridge regression

Simple algebra tells us that

$$\widehat{\beta} - \beta_0 = -\lambda \widehat{V}_\lambda^{-1}\beta_0 + \frac{1}{n}\widehat{V}_\lambda^{-1}X^\top \epsilon =: b + v$$

Analysis of the bias term. Note that

$$\left|x^\top \widehat{V}_\lambda^{-1}\beta_0\right| \le \sqrt{x^\top \widehat{V}_\lambda^{-1}x} \cdot \sqrt{\beta_0^\top \widehat{V}_\lambda^{-1}\beta_0}.$$

With $\|\beta_0\|_2 \le B$ and $\widehat{V}_\lambda^{-1} \le 1/\lambda$, we have that

$$\left|\lambda x^\top b\right| \le \sqrt{\lambda} \cdot \sqrt{x^\top \widehat{V}_\lambda^{-1}x}.$$

Analysis of the variance term. We have $v \sim \mathcal{N}(0, n^{-1}\widehat{V}_\lambda^{-1}\widehat{V}\widehat{V}_\lambda^{-1})$. So for any fixed $x \in \mathrm{supp}(P)$, with high probability

$$\left|x^\top v\right| \lesssim \sqrt{\frac{x^\top \widehat{V}_\lambda^{-1}\widehat{V}\widehat{V}_\lambda^{-1}x}{n}} \le \sqrt{\frac{x^\top \widehat{V}_\lambda^{-1}x}{n}},$$

where the last inequality holds because $\widehat{V} \preceq \widehat{V}_\lambda$. Subsequently,

$$\left|x^\top(\widehat{\beta} - \beta_0)\right| \le (\sqrt{\lambda} + 1/\sqrt{n})\sqrt{x^\top \widehat{V}_\lambda^{-1}x} =: \Delta(x).$$

Consequently

$$\mathbb{E}[\Delta(x)|P] = \left(\frac{1}{\sqrt{n}} + \sqrt{\lambda}\right)\mathbb{E}[\sqrt{x^\top \widehat{V}_\lambda^{-1}x}|P]$$

For two symmetric matrices A, B, we write $A \preceq B$ if $x^\top Ax \le x^\top Bx$ for all $x \in \mathbb{R}^d$. We write $A \succeq B$ if and only if $B \preceq A$.

Proposition 0.1. *If $A \preceq B$ then for any square matrix D, $DAD^\top \preceq DBD^\top$.*

Proof. For any $x \in \mathbb{R}^d$, $x^\top(DAD^\top)x = z^\top Az \le z^\top Bz = x^\top(DBD^\top)x$, where $z = D^\top x$. Note that this is true regardless of whether A or B is positive semi-definite. □

Proposition 0.2. *For every positive definite A and B, $A \preceq B$ implies $A^{-1} \succeq B^{-1}$.*

Proof. Because $A \preceq B$, the previous proposition implies that $C = B^{-1/2}AB^{-1/2} \preceq I$. Since A is pd, $B$is inevrtible, C is also pd and therefore all eigenvalues of C are between 0 and 1. This means all eigenvalues of C^{-1} are between 1 and $+\infty$, implying $C^{-1} \succeq I$. This means $B^{1/2}A^{-1}B^{1/2} \succeq I$ and invoking the previous proposition again we have $A^{-1} \succeq B^{-1}$. □

Lemma 0.1. *If $\lambda \gtrsim B^2 \log n/n$ then with high probability $0.5V_\lambda \preceq \widehat{V}_\lambda \preceq 1.5V_\lambda$.*

Proof. Define $z_i := V_\lambda^{-1/2}x_i$. Because $V_\lambda \succeq \lambda I$ and $\|x_i\|_2 \le B$ almost surely, we know that $\|z_i\|_2 \le B/\sqrt{\lambda}$ almost surely. Note also that

$$\mathbb{E}[z_i z_i^\top] = V_\lambda^{-1/2}\mathbb{E}[x_i x_i^\top]V_\lambda^{-1/2} = V_\lambda^{-1/2}VV_\lambda^{-1/2} = I - \lambda V_\lambda^{-1}.$$

Because $\|V_\lambda^{-1}\|_{\mathrm{op}} \le \lambda^{-1}$, we have that $\|\mathbb{E}[z_i z_i^\top]\|_{\mathrm{op}} \le 1$. Invoking [? , Lemma 1], we have

$$\Pr\left[\left\|\frac{1}{n}\sum_{i=1}^n z_i z_i^\top - \mathbb{E}[z_1 z_1^\top]\right\|_{\mathrm{op}} > t\right] \le 2\min\{d,n\}^2 \cdot e^{-\frac{n(t-1)\lambda}{4B^2}}.$$

Subsequently, with high probability

$$\left\|\frac{1}{n}\sum_{i=1}^n z_i z_i^\top - \mathbb{E}[z_1 z_1^\top]\right\|_{\mathrm{op}} \lesssim \frac{B^2 \log n}{\lambda n} \le \frac{1}{2},$$

where the last inequlity holds if $\lambda \gtrsim B^2 \log n/n$. Note that $\frac{1}{n} z_i z_i^\top = V_\lambda^{-1/2} \widehat{V} V_\lambda^{-1/2}$. Subsequently, the above inequality implies

$$\frac{1}{2} I - \lambda V_\lambda^{-1} \preceq V_\lambda^{-1/2} \widehat{V} V_\lambda^{-1/2} \preceq \frac{3}{2} I - \lambda V_\lambda^{-1}.$$

Using Proposition 0.1, we have

$$\frac{1}{2} V_\lambda - \lambda I \preceq \widehat{V} \preceq \frac{3}{2} V_\lambda - \lambda I.$$

Therefore,

$$\frac{1}{2} V_\lambda \preceq \widehat{V}_\lambda \preceq \frac{3}{2} V_\lambda.$$

□

Now, using the lemma and Proposition 0.2, we have that

$$\begin{aligned}
\mathbb{E}[\Delta(x)|P] &= \left(\frac{1}{\sqrt{n}} + \sqrt{\lambda}\right) \mathbb{E}\left[\sqrt{x^\top \widehat{V}_\lambda^{-1} x}|P\right] \\
&\le 2\left(\frac{1}{\sqrt{n}} + \sqrt{\lambda}\right) \mathbb{E}\left[\sqrt{x^\top V_\lambda^{-1} x}|P\right] \\
&\le 2\left(\frac{1}{\sqrt{n}} + \sqrt{\lambda}\right) \sqrt{\mathbb{E}[x^\top V_\lambda^{-1} x|P]} \\
&= 2\left(\frac{1}{\sqrt{n}} + \sqrt{\lambda}\right) \sqrt{\langle V_\lambda^{-1}, V\rangle} \\
&= 2\left(\frac{1}{\sqrt{n}} + \sqrt{\lambda}\right) \cdot \sqrt{\sum_{i=1}^d \frac{\mu_i}{\mu_i + \lambda}},
\end{aligned}$$

where $\mu_1, \cdots, \mu_d$ are eigenvalues of $V = \mathbb{E}[xx^\top]$.

Note: this analysis requires $\mathbb{E}[\varepsilon|X] = 0$; that is, full exogeneity.

0.3 Emprirical process analysis

There are many ways to construct the basic inequality, either from first-order KKT conditions or from minimizers of least squares. We are going to start from

$$(\widehat{\beta} - \beta_0)^\top V (\widehat{\beta} - \beta_0) \le 2\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i x_i^\top (\widehat{\beta} - \beta_0)\right|.$$

For any vector ϕ let $\|\phi\|_K := \sqrt{\phi^\top K \phi}$. Then

$$\|\widehat{\beta} - \beta_0\|_V \leq 2 \sup_{\phi : \|\phi\|_V \leq 1} \frac{1}{n} \sum_{i=1}^n \varepsilon_i x_i^\top \phi \lesssim \sqrt{\frac{d}{n}},$$

where the second inequality is from concentration inequalities of self-normalized empirical process. Consequently, for any x,

$$\left| x^\top (\widehat{\beta} - \beta_0) \right| \leq \|x\|_{V^{-1}} \|\widehat{\beta} - \beta_0\|_V \lesssim \sqrt{\frac{d x^\top V^{-1} x}{n}} =: \Delta(x).$$

If you compare this with the $\Delta(x)$ in the previous section, you'll notice an extra $\sqrt{d}$ factor. This leads to $\Psi = \widetilde{O}(d/\sqrt{n})$ rather than $\widetilde{O}(\sqrt{d/n})$.

With this extra d factor, there are two good things about this argument:

1. It holds almost surely for any x.
2. It only requires $\mathbb{E}[\varepsilon_t | x_t] = 0$ so that $\sum_{i=1}^n \varepsilon_i x_i^\top \phi$ is a martingale difference sequence for any ϕ. So x_t can depend on ε_{t-1}.
3. It is relatively easier to convert "on-average" error bounds to "pointwise" ones, by using the last Cauchy-Schwarz step.

References